Packaging Guidelines

› Packaging Guidelines Rev01

This document is controlled! Please refer to the Kenosha Sharepoint Digital Training library for the latest version.

OBJECTIVE

We want to use appropriate packaging to ensure our customer's parts arrive in the same condition they were shipped.

1. Commitment to doing it right the first time
2. Attention to detail
3. Protect the product as if it were our own
4. Reference your Standard Work Instructions (SWIs).
5. One order at a time.
6. Work from Bin to finished box
7. Follow all packaging guidelines
8. Give special attention to TX5 items, such as motherboards, LCD screens, memory components, etc.…
9. ESD procedures and packaging must be used for all electronic parts

Never forget that the next person to see your packaging is your customer!

What we'll learn today?

- Mandatory Rules for Packaging
- Definitions
- ESD
  - ESD: Anti-Static/ Static Shielding Bags
- Carton Packaging
- Filling the Voids
- Mixed GPN Packaging
- Media Packaging
  - Motherboards
  - Memory
  - Hard Drives
- Taping and Sealing

- General Pallet Rules
- Pallet Requirements
- Pallet Dimensions & Weights
- Mixed Pallets
- Pallet Double Stacking Rules
- International Palletized Loads
- Plastic Stretch Wrap
- Double Stacking Rules

MANDATORY RULES

1. All parts should be packed in their designated packaging.
2. Any part(s) packed in a bag or box is considered a case.
3. No International shipments can be shipped without approval from the International Clerks.
4. Apart from racks and other oversized components, all parts must be inside an appropriate package with appropriate dunnage.
5. Manufacturer packaging should be used when possible and when boxes don't contain many non-required labels.
6. Parts shipped in less than a full box quantity must be secured in a packing box with brown packing paper.
7. Reused corrugated material is acceptable as long is not damaged.
8. No product should be packed in an open-faced or display box.
9. Parts with exposed circuit boards cannot directly contact other parts. They are to be in their slot/insert or wrapped and sealed in red bubble wrap.
10. Pallet loads into Zebra sites must comply with pallet requirements.

DEFINITIONS

- **Pack Consolidation Report (PCR):** Order details and packer instructions.
- **Partial order:** is when a pick is made from a master pack of a product, and it does not entirely consume the pack's quantity
- **Full box quantity:** A package or container that hasn't been picked from or opened.
- **ESD package:** Packaging to control the sudden flow of electricity between two electrically charged objects
- **Electrostatic Discharge (ESD) Materials:** Materials intended to reduce or otherwise redirect static electricity from contacting sensitive electronic devices. ESD materials are typically black or pink in color.
- **Standard Pallet:** L48" x W40" wooden/black plastic, 4-way pallet
- **Heat Treated (HT) Pallet:** For international shipments only.
- **Plastic Pallet:** Brazil & China shipments only.
- **Oversized Item:** An item that does not fit on a standard-sized pallet
- **COO (Country of origin):** the country where a product is manufactured or produced or the final location where a "substantial transformation" of the imported components occurs.
- **A Case:** An item packed in a box or a bag.
- **Quality Check (QC) Traveler:** A checklist to ensure proper steps were followed.

Pack Consolidation Report (PCR)

Pack consolidation report for delivery:

Delivery Number:
79060568

TX4

Customer:
Advantage Label and Packaging

Packer Initials:

Ship To:
Attn: Josh Ambrose
Byrne Electrical Specialists
320 Byrne Industrial Drive
Rockford,MI 49341
United States

Ship method:
BESTWAY GROUND

Original Ship method:
BESTWAY GROUND

Planned Ship Date:
31-JUL-2023

Freight Terms:
Prepay & Add

Inco Terms:
FCA

Consolidation locator:
SA.201.A.00.

Final Destination:
ROCKFORD, US

Nr.	LPN	Length	Width	Height	Weight	Cartons
1	TX40016401					

Item	Order Qty	LPN	Picked Qty	Order UOM	Comments
P1004234	2	TX40016401	2	EA	

Pack consolidation report for delivery:

Delivery Number:
83398698

TX4

Customer:
Ingram Micro Data Capture Pos Div.

Packer Initials:

Ship To:
Ingram Micro Data Capture Pos Div.
PO Box 25189
Santa Ana,CA 92799
United States

Ship method:
FEDEX 2DAY

Original Ship method:

Planned Ship Date:
11-MAR-2024

Freight Terms:

Inco Terms:

Consolidation locator:
SA.001.A.00.

Final Destination:
SANTA ANA, US

Batteries Exception ___

Nr.	LPN	Length	Width	Height	Weight	Cartons
1	TX40016788					
2	TX40016789					

Item	Order Qty	LPN	Picked Qty	Order UOM	Comments
ET40AA-001C1B0-NA	1	TX40016789	1	EA	DG:UN 3481 LITHIUM ION BATTERIES CONTAINED IN EQUIPMENT 9 SECTION I
ZT61042-T010100Z	1	TX40016788	1	EA	DG:NO LABELING REQUIREMENT FOR AIR TRANSPORT

Pack Consolidation Report (PCR)

Pack consolidation report for delivery:

Delivery Number:
83393287

TX4

Customer:
FEDEX EXPRESS CORP

Packer Initials:

Ship To:
FEDEX EXPRESS CORP
3690 HACKS CROSS RD
BLDG I::3RD FL:
Memphis,TN 38125
United States

Ship method:

Original Ship method:

Planned Ship Date:
11-MAR-2024

Freight Terms:

Inco Terms:

Consolidation locator:
SA.001.A.00.

Final Destination:
MEMPHIS, US

Batteries Exception ___

Nr.	LPN	Length	Width	Height	Weight	Cartons
1	TX40016787					

Item	Order Qty	LPN	Picked Qty	Order UOM	Comments
BTRY-MPP-68MA1-01	1	TX40016787	1	EA	DG:LABEL CARTON AS CARGO AIRCRAFT ONLY. UN 3480 LITHIUM SECTION IB

Pack consolidation report for delivery:

Delivery Number:
83393288

TX4

Customer:
Boreal Argentina

Packer Initials:

Ship To:
BOREAL TECHNOLOGIES SA
11 DE SEPTIEMBRE 4717
FL PISO 6A:NUNEZ
BUENOS AIRES,CF
Argentina

Ship method:

Original Ship method:

Planned Ship Date:
11-MAR-2024

Freight Terms:

Inco Terms:

Consolidation locator:
SA.001.A.00.

Final Destination:
BUENOS AIRES, AR

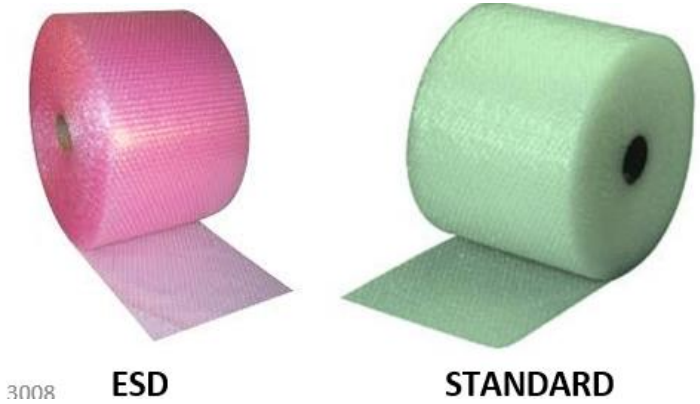
Batteries Exception ___

Nr.	LPN	Length	Width	Height	Weight	Cartons
1	TX40016790					

Item	Order Qty	LPN	Picked Qty	Order UOM	Comments
BAT-001-00	1	TX40016790	1	EA	DG:LABEL CARTON AS CARGO AIRCRAFT ONLY. UN 3090 LITHIUM SECTION IB

ESD

- Non-factory sealed ESD items are isolated and boxed separately from non-ESD.
- ESD GPNs should not be mixed in the same box.
- All static-sensitive parts are packaged in ESD bags and trays.
- Use separators, covers, and foam inserts as appropriate for the packaged commodity.
- Boxes are lined with 2-inch brown packing paper before placing products in the box.
- Use sufficient dunnage (at least 1-2 layers of ESD bubble wrap, if needed or non-ESD) to protect the part.
- Shipping Motherboards in ESD bags does not prevent damage; therefore, the boards must be individually wrapped in pink bubble wrap or packaged in the appropriate box with isolated inserts.



ESD Packaging

ESD – Bags

- Silver translucent Static Shielding Bags are Static Protective made of a multi-layer material that prevents static electricity build-up.
- Silver bags protect sensitive electronic components from static shocks when handled.
- Silver bags you cannot see through are Static and Moisture Protective.
  - **Never** staple or otherwise puncture a static shielding bag! Take any additional precaution to pad sharp corners present on the part(s) that may puncture the bag while in transit.
  - Static shielding bags should not be reused. If a bag is wrinkled/creased, the shield is broken and cannot protect the part.

- Pink Poly bags are static dissipative
- Clear Static Dissipative Bags are unsafe for components, as a component may absorb an ESD shock. Hence, only non-ESD sensitive products render safe to use with static dissipative bags

Silver Static Shielding Bag

Pink Poly Static Dissipative

CARTON PACKAGING

1. Ensure item(s) are manageable for the selected shipping carton.

- Ensure item(s) fill ~80% of the selected shipping carton. If >20% is not filled, select a smaller shipping carton.
- If a smaller carton is not available, fill it with brown packing paper as needed to fill the shipping carton completely
- Small non-ESD parts such as screws, plastic pieces, and small metal pieces are approved to ship in padded mailers for a more optimal shipping experience

2. Line the Shipping Carton with 2 inches of brown packing paper to protect the bottom and all four sides of the carton.

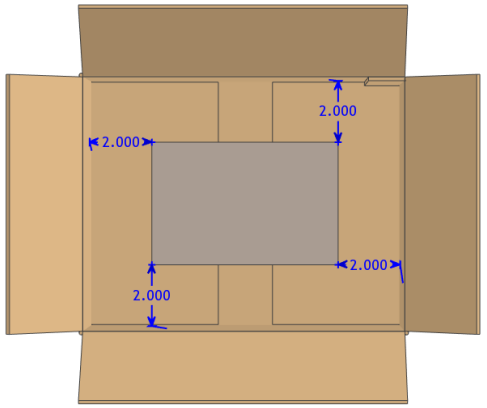
- Criss-cross wrapping method across top closure
- Material in the inner box must be packed tightly to minimize movement during shipping. The movement inside the box usually results in damage
  - Contents should always be centered within the container
  - Under-filled boxes are likely to collapse
  - Overloaded boxes may burst
  - It is better to break down your shipment into two or more cartons than have one very heavy carton

3. All cartons and packaging surfaces must be flat and robust to support the package weight without distortion, deflection, or bending.

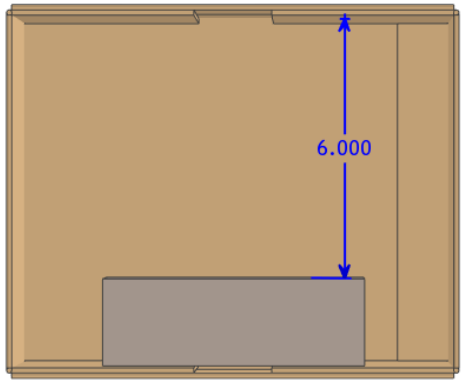
- Cartons must be free of any punctures, tears, or damage

FILLING THE VOIDS

- Any leftover space in the carton should be filled with brown packing paper. Ensuring no damage occurs during transport.
- There should be at most 2 inches (5 cm) of space between the part and the side of the container and at most 6 inches (15 cm) between the part and the top of the container.
  - If exceeded, use a smaller carton
  - If a smaller carton is not available, fill it with brown packing paper as needed to completely fill the shipping carton.



2-inch (5cm) maximum between product and side of container



6-inch (15cm) maximum between the product and the top of the container

MIXED GPN PACKAGING

- Mixed GPNs are not allowed per carton unless each GPN is in an individual box with individual labels inside the overpack
- Components should be secured in their own packaging compartment (tray, foam insert, etc).
- To help visually determine if an overpack has mixed "**MIXED PARTS**" label must be applied to the overpack.
- Outer packaging for cartons should follow general shipping carton requirements listed on Carton Packaging slide 9

Pack and Seal Shipping Carton:

- Tape should be wrinkle-free and must extend 2 inches (5cm) on the top sides of the box
- Taping should follow Taping and Sealing slides 15-16.

MEDIA PACKAGING – MOTHERBOARDS, LARGE LCDs

****Do not break vendor-sealed boxes to consolidate material**.**

****Reuse the foam the item was packed with from the supplier**.**

1. Remove top foam from corrugated box

2. Insert product into the box (product must be bagged in a static shielding bag as shown)

3. Replace top foam, flat side should face away from the product

4. Seal the box according to the Taping and Sealing Method slides 13-14

MEDIA PACKAGING - MOTHERBOARDS – MULTI

****Do not break vendor-sealed boxes to consolidate material**.**

****Reuse the foam the item was packed with from the supplier**.**

1. Select appropriate shipping carton.

2. Use carton slots if available, if not available use the brown packing paper as separator.

3. Use brown packing paper and/or original supplier foam if available.

4. Seal the box according to the Taping and Sealing Method slides 13-14

TAPING AND SEALING – 'H' METHOD

- Use the H-taping method to seal the outer container using the required tape
- Before filling the box with your items, tape the bottom of the box first to provide reinforcement and prevent it from collapsing when lifted
- Apply pressure on the whole length of tape to ensure it adheres to the cardboard box. This will prevent moisture from getting inside and tampering and ensure box security.
- Overpack Parcel shipments H-method taping is to be used.

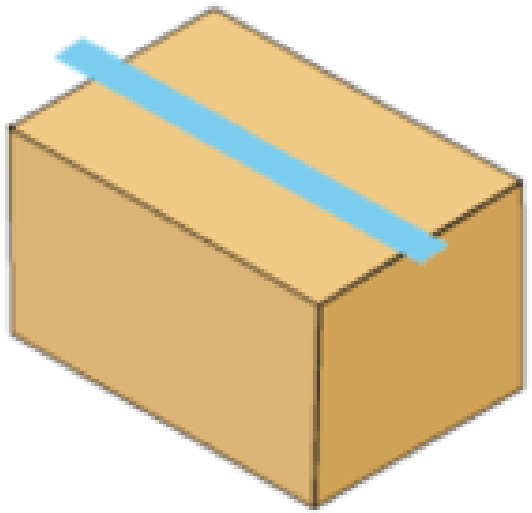
1. Fold outer flaps and tape down middle of the bottom of the box. Provide 2" overhang on each end.

2. Tape down sides of the bottom of the box. Provide 2" overhang on each end.

3. Flip the box. Add products, dunnage, etc. Repeat Steps 1&2 for top side of the box.

TAPING AND SEALING – 'I' METHOD

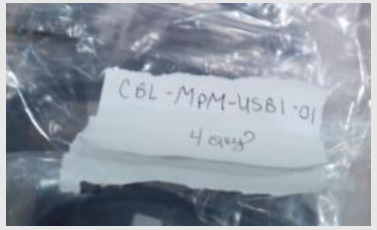
- For LTL shipments with overpack boxes, a single strip of tape with 2” overhang will suffice.
- Apply pressure on the whole length of tape to ensure it adheres to the cardboard box.



Remember to leave a 2” overhang on each end

LABELS – GENERAL RULES

1. Old shipping labels and all indicators must be removed from any reused packaging; if the label cannot be removed, apply a brown packing label.
2. Labels should not obstruct any handling symbols.
3. Leave sufficient space from the edges for edge boards, corner boards, and banding
4. Labels should not cover other relevant labels, such as part information.
5. Labels should be vertically straight and parallel with the sides of the carton or pallet
6. The label should be placed as high as possible on the pallet
7. Labels must be affixed to one side where all labels are visible
8. Labels should be applied wrinkle-free as barcode labels may not be scannable
9. Labels should not be placed over a seam, under the sealing tape, the stretch wrap, or banding
10. Labels should not be placed in a manner that interferes with the scanning of the label
11. Absolutely NO handwritten labels of any kind!

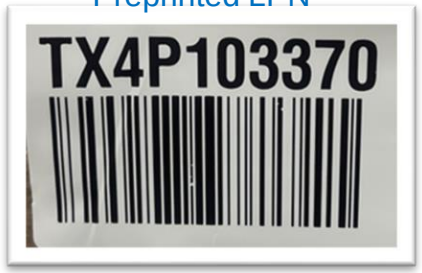


LABELS – GENERAL RULES

Shipping Contents Label

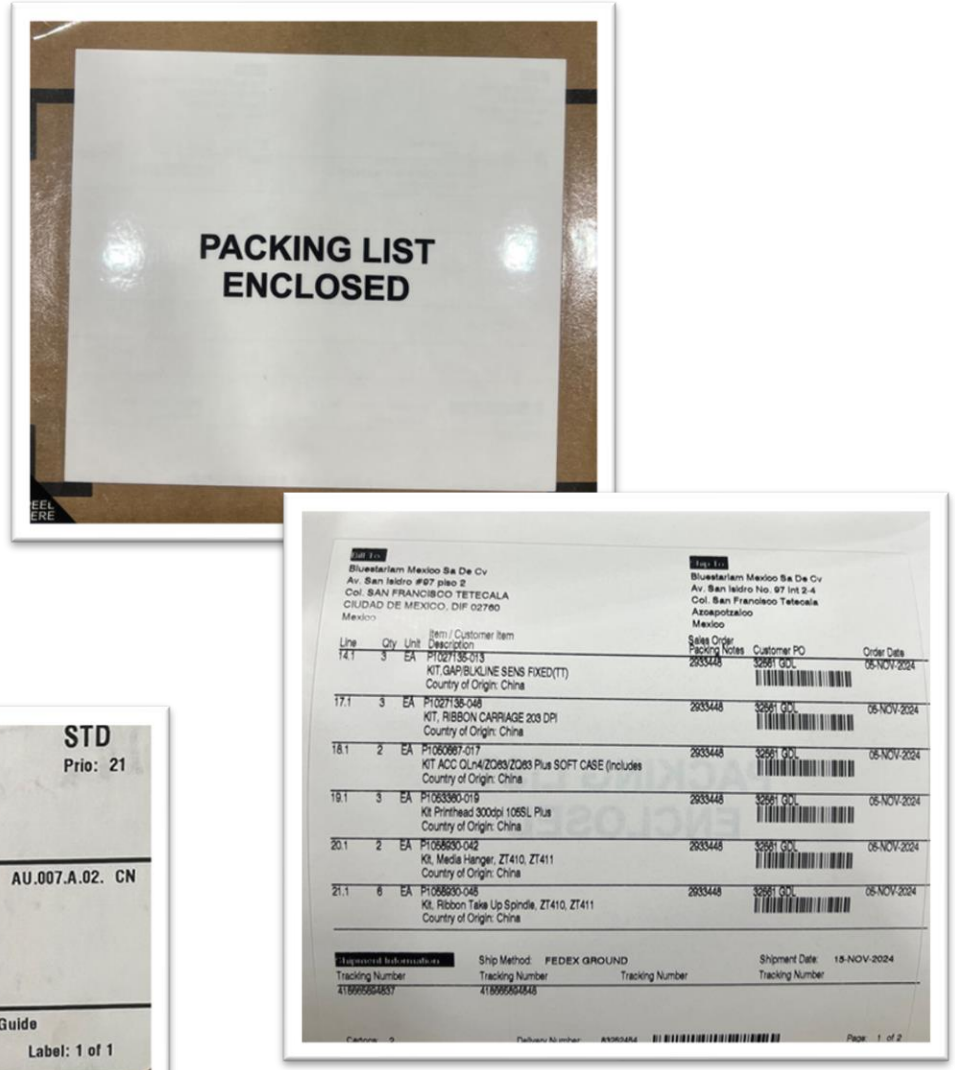
LPN Labels

Preprinted LPN



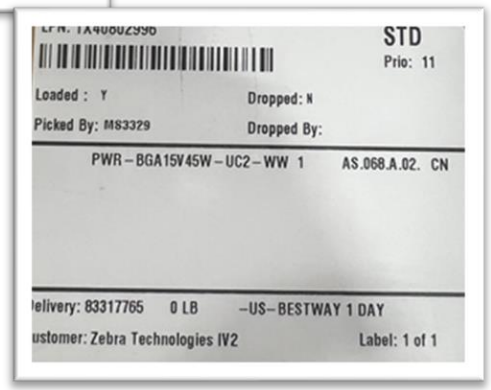
Print on Demand LPN

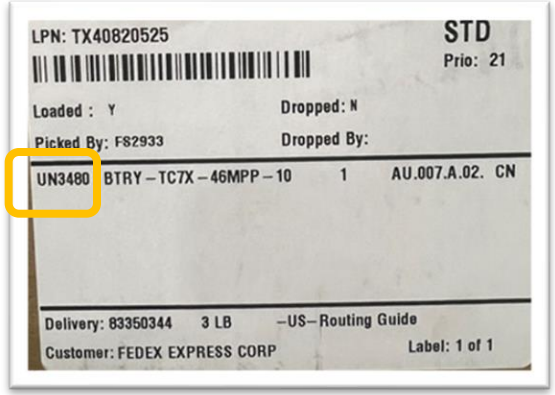
zSlip



Pick Drop Audit Labels

Non-DG Drop Audit Label

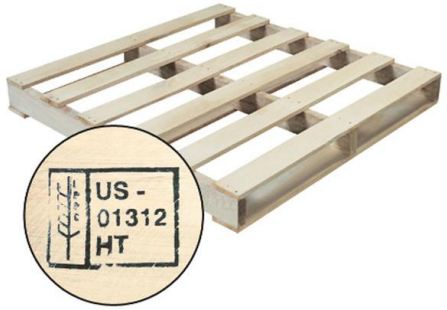




PALLET REQUIREMENTS

- **Acceptable Pallet types:**
  - **Globally Acceptable Pallet:**
  - Block Pallet (1200 x 1000mm / 48” x 40”) GPN 1033200 - Must be 4 sided accessible with full perimeter bottom deck-boards
  - **Regional Pallet Options:**
    - US shipments only: NA Standard Notched Stringer Pallets est.(1219 x 1016mm / 48” x 40”) - Must be 4-sided accessible with full perimeter bottom deck-boards
  - Wood or Plastic Only; New or recycled (if they still comply with all other pallet requirements)
  - Wood pallets should be made of high-quality lumber
  - Long load pallets should also be four sides accessible with full perimeter bottom deck-boards
  - Long loads should be palletized according to the material need. Long loads are typically oversized items that cannot be palletized on a standard 48x40 footprint pallet.
  - Block Style Pallets: All blocks must be present.

✓ Acceptable stringer pallet



✓ Acceptable Heat-Treated pallet

✓ Acceptable plastic pallet

✓ Acceptable block pallet

GENERAL PALLET RULES

- 1” inch overhang is allowed when building a pallet.
- Pallets must be in new or good condition with bottom deck boards present and not damaged.
- Minor cracks are allowed if the board is secured to the stringer.
- All cartons must be secured to the pallet using plastic stretch wrap.
- Pallets must have the proper Heat-Treated (HT) stamps and be used for International shipments.
- Plastic pallets are for shipments to China and Brazil.

- **Unacceptable Pallets:**
  - Pallets without bottom and top deck boards
  - Missing 25% part of the top or bottom deck boards.
  - Cracks in the stringer.
  - Pallets that are not 4-way accessible
  - Pressed board honeycomb, paper, Pressed wood, Metal, Foam

Unacceptable

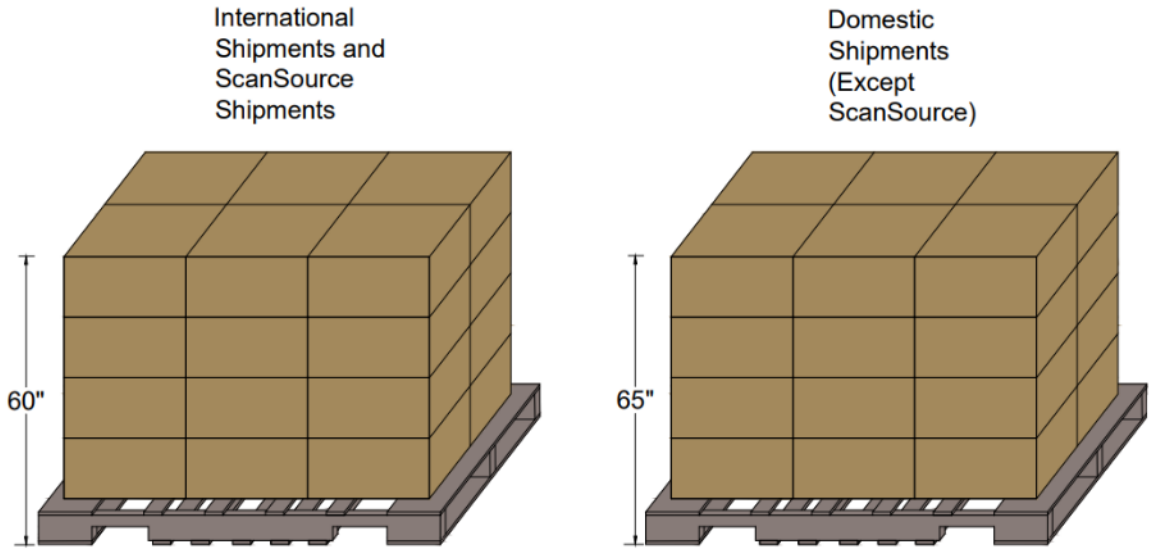
Acceptable

PALLET DIMENSIONS AND WEIGHTS

- **Fork pocket openings should be:**
  - Minimum 3.75” inside pocket height
  - Minimum 31” inside pocket width
- **Height is the total height from floor to top of the load.**
  - Max Pallet Height for National shipments Including Pallet: 65” [1651mm]

    Except for ScanSource, UPS, Walmart and International shipments.

  Min Pallet Height Including Pallet: 19.6” [500mm]
  - Maximum weight: 2149 lbs. [975 Kg]



CUSTOMER PALLET REQUIREMENTS

- **BLUESTAR**
  - Height: 65" max
  - Overhang: 1” max

- **USPS**
  - Height: 48" max
  - Overhang allow? No

DOMESTIC ORDERS
Height: 65” max
Overhang: 1” max

INGRAM
Height: 65” max
No block pallets

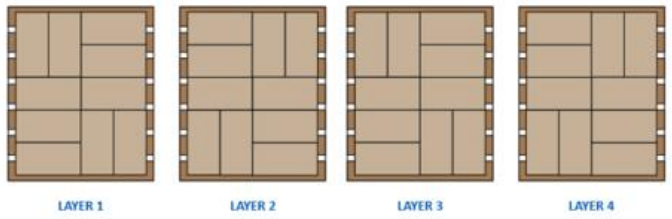
WALMART
Height: 58” max
Overhang: 1” max

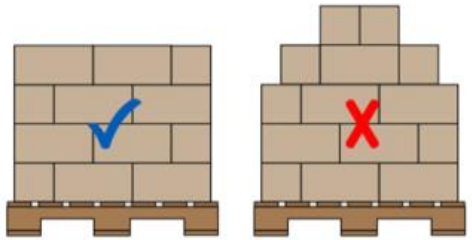
INTERNATIONAL ORDERS
Height: 60” max
Pallets: Heat Treated
Brazil & China – Plastic pallets
Overhang: 1” max

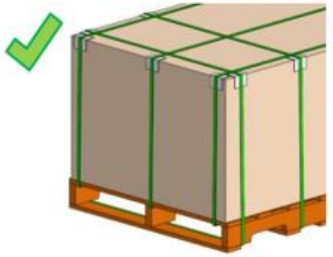
SCANSOURCE
Height: 60” max

MIXED PALLETS

1. All items must be secured to the pallet
2. The heaviest items must be placed on the bottom layer of the pallet
3. Boxes should be stacked on a pallet in an interlocking pattern, when possible, as shown in images
4. Arrange boxes in a way that brings them as close to the edges of the pallet as possible
5. Never pyramid stack the boxes
6. No objects should obscure the fork pocket area of the pallet
7. A 1-inch overhang is allowed in the build of the pallet
8. Boxes should always be shipped in their proper shipping orientation
9. Boxes should never be shipped on their side; check box printing for side up arrows and keep the arrows pointing up or any other specific instructions printed on the boxes







PALLETIZED LOADS

Steps:

1. Pallet loads should be covered with plastic wrap to cover the top of the load.
2. Then; the load should be stretched and wrapped in a white wrap.

Quality Check Traveler

Before a pallet is moved to the wrapping station, it must be pre-audited to ensure proper steps are followed before the quality inspection.

CONFIDENTIAL INFORMATION

Quality Check (QC) Traveler

Case Count: _____ **Packer:** _____ **Pre-Auditor:** _____ **Manifester:** _____

Did the case count change? If YES, it must be updated in the RF. Full Name Full Name Full Name

- ○ **Follow PCR Notes**: PCR Notes contain vital information for shipment processing.

- ○ **Acceptable pallets** used for the shipment (free of damage).
  - Heat Treated for international shipments.
  - Plastic for Brazil and China shipments.
  - Ingram Macro NO Block Pallets

- ○ **Contents:**
  Correct > SKU#, Qty & Serial #s.

- ○ **Dangerous Goods:** DG correctly identified & adequately labeled. *Reference DG posted Job Aids.*
  UN3480 or UN3090 Section IA **(AIR)** - shipment MUST have 3 DG labels & Shippers' Declaration
  • Hazard label (Class 9) • Cargo Aircraft Only (CAO) label • Shipper/Consignee net wt. label.

  UN3480 or UN3090 Section IA **(GROUND)** - shipment MUST have 3 DG labels.
  • Hazard label (Class 9) • CAO label • Shipper/Consignee net wt. label

  Section IA Batteries must be in a UN Spec Box.

  UN3480 or UN3090 Section IB **(AIR)** - shipment MUST have 4 DG labels & Shippers' Declaration
  •Battery label • CAO label • Hazard label (Class 9) • Shipper/Consignee net wt. label.

  UN3480 or UN3090 Section IB **(GROUND)** - shipments MUST have 2 DG labels.
  •Battery Label, • CAO label

  UN3481 Section I:
  - **AIR** – Class 9, Shipper/Consignee label & Shippers' Dec.
  - **GROUND** - Class 9 & Shipper/Consignee label.

  UN3481 or UN3091 Section II **(AIR or GROUND)** – - shipments need 1 DG label. The Battery Label

- ○ **Workmanship:**
  Boxes are free of damage and markings—NO handwritten labels and correct label placement.
  - Does your pallet need Corner boards? ⇨USPS must have corner boards.

- ○ **Labeling:**
  Parcel – Shipping Contents label for more than 2-part items.
  LTL:
  - Part label affixed to non-vendor boxes.
  - Shipping Detailed Content Label & Shipping Contents Label on each pallet.

- ○ **Packaging:**
  Follow vendor-sealed box orientation, proper ESD packaging utilized, and sufficient dunnage used.

- ○ **Pallet Height:**
  The overhang must be less than 1 inch. ** USPS cannot have any overhang.
  ❖International and ScanSource max height 60 inches
  ❖ Domestic 65 inches or less
  ❖Walmart pallets cannot exceed 58 inches.
  ❖ USPS cannot exceed 48 inches.

- ○ **Paperwork:**
  Correct BOL & Packing List
  International shipments must have the Commercial Invoice. Ensure the COO is present and not XX.
  Brazilian shipments must have a Zebra blue stamp and blue signature on the invoice and packing slip.

PLASTIC STRETCH WRAP

1. Stretch wrap must extend low enough to secure the cartons to the pallet adequately
2. Wrap shall not entirely obscure any fork opening
3. Stretch wrap must provide adequate tension to prevent any shifting beyond the pallet edge
4. Excess stretch wrap must not extend past the outer footprint of the pallet
5. The stretch wrap should not cover the IPPC Heat Treatment Stamp. The stamp should always remain visible.
6. Stretch wrap should be tied to the pallet
7. Automated stretch wrap machines should be utilized when available

PALLET DOUBLE-STACKING RULES

General Rules:

- Do not double stack unless otherwise approved by the Zebra Leadership Team.
- Generally approved for empty packaging shipments if the packaging is designed for double stacking and/or if otherwise approved by the Zebra Leadership Team.
- Should be indicated with a ‘no stack pallet cone’ and/or "Do Not Double Stack" sticker, as shown below, excluding international air shipments.
- Do not shrink wrap or strap to bind top and bottom pallets. This is to avoid unloading concerns and/or if there are incidents such as fire, damage, or weak bottom pallets.
- Pallets are loaded onto the trailer with a pallet jack, walkie-ryder, and stand-up forklift truck.

Revision History

Rev.	Rev. Date	Description of Change	Revised By
01	1/24/2023	Original issue	Adriana Cabrera